4-22 What is the range of the gain v_O/v_S in Figure P4–22?

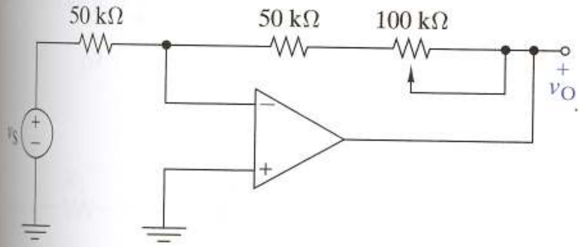


FIGURE P4 – 22

4-23 (a) Find v_O in terms of v_S in Figure P4–23.
(b) Find i_O for $v_S = 1.5$ V.

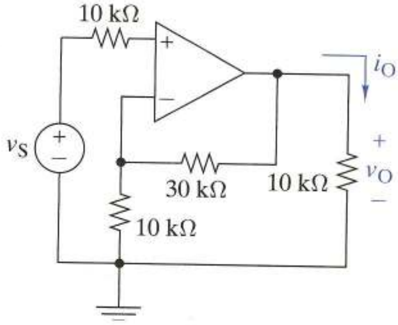


FIGURE P4 – 23

4–25 What is the gain v_O/v_S in Figure P4–25 when the switch is in position 1? Repeat for positions 2 and 3.

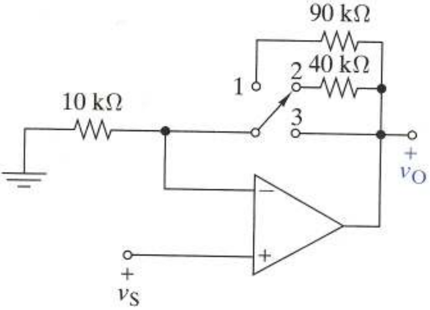


FIGURE P4 – 25

4–26 Find v_O in terms of the inputs v_1 and v_2 in Figure P4–26.

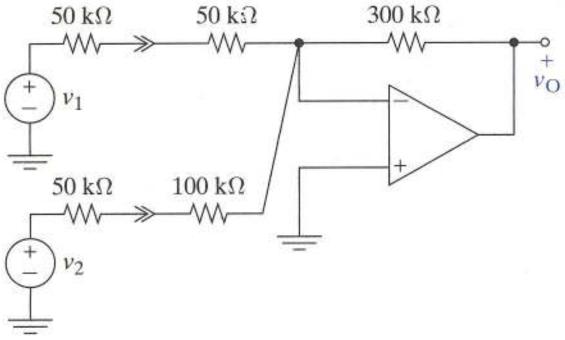


FIGURE P4–26

4–30 Find v_O in terms of the inputs v_{S1} and v_{S2} in Figure P4-30

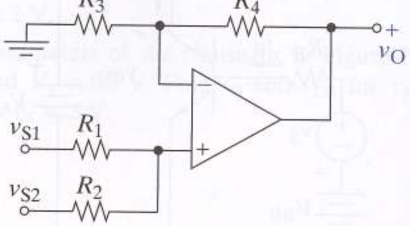


FIGURE P4–30

4–34 Use node-voltage analysis in Figure P4–34 to show that $i_O = -v_S/2R$ regardless of the load. That is, show that the circuit is a voltage-controlled current source.

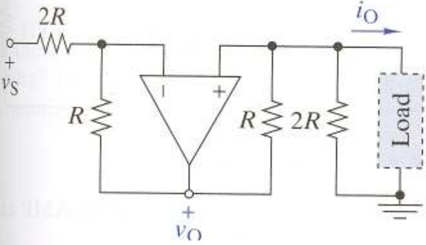
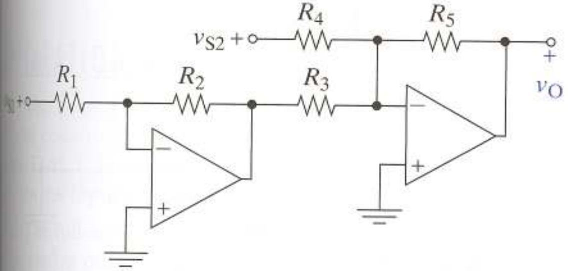


FIGURE P4–34

–36 Find v_O in terms of the inputs v_{S1} and v_{S2} in Figure P4–36.



IGURE P 4 – 36

4–37 Find the output v_2 in terms of the input v_1 in Figure P4–37.

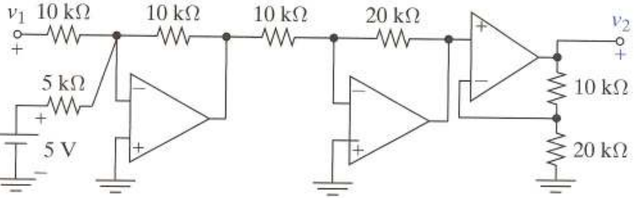


FIGURE P4 – 37

4–47 **D** Design the interface circuit in Figure P4–47 so that the output is $v_2 = (50v_1 + 3)$ V.

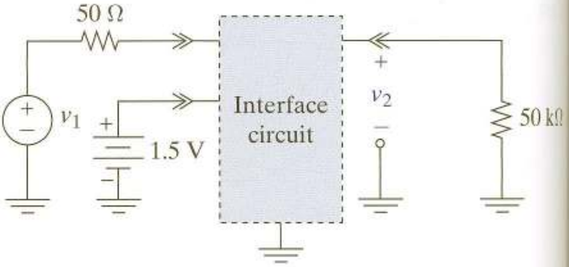


FIGURE P4 – 47

4–55 **D** A medical-grade pressure transducer has been developed for use in invasive blood pressure monitoring. The output voltage of the transducer is $v_{TR} = (0.05P - 0.75)$ mV, where P is pressure in mmHg. The output resistance of the transducer is 300Ω. The blood pressure measurement is to be an input to an existing multisensor monitoring system. This system treats a 1-V input as a blood pressure of 20 mmHg and a 10-V input as 200 mmHg. Design an OP AMP circuit to interface the new pressure transducer with the existing monitoring system.